Woollen clothes keep us warm in winter. It is just because woollen clothes have fibres and between those fibres, air is trapped which reduces heat loss. Air reduces heat loss because it is an insulator or poor conductor of heat. Thus statement (A) is correct but reason (R) is false.

93. Which one of the following statements is not correct?

(a) The velocity of sound in air increases with the increase of temperature

(b) The velocity of sound in air is independent of pressure

(c) The velocity of sound in air decreases as the humidity increases

(d) The velocity of sound in air is not affected by the change in amplitude and frequency

I.A.S. (Pre) 2003

Ans. (c)

Humidity is the percentage of water vapour present in air. As the humidity increases, the percentage of water vapour in air increases and this decreases the density of air. This results in the increase of velocity of sound. So, increase in the humidity of air increases the velocity of sound in air. Thus, statement (c) is not correct while statements of other three options are correct.

94. Opening the door of refrigerator kept in the room –

(a) You can cool the room to some degree

(b) You can cool the room to the temperature of refrigerator

(c) You can warm the room a little

(d) You can neither cool nor warm the room

39th B.P.S.C. (Pre) 1994

Ans. (c)

If the door of a refrigerator kept open in a room, the temperature will start to rise inside the refrigerator. The thermostat will kick in and try to cool it back down. This means the motor is running, which means heat is being added to the room resulting warmer room.

95. If the doors of a refrigerator are left open for few hours, then the room temperature will :

(a) decrease

(b) increase

(c) remain the same

(d) decrease only in the area in the vicinity of the refrigerator

(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (b)

See the explanation of above question.

Wave Motion

Notes

Wave :

- Wave involves the transfer of energy without the transfer of matter. In conclusion, a wave can be described as a disturbance that travels through a medium, transferring energy form one location (its source) to another location without transfer of matter.
- Waves are of two types :
  1. Mechanical Waves.
  2. Electromagnetic Waves.

1. Mechanical Waves :

- Mechanical wave is a wave that is an oscillation of matter, and therefore transfers energy through a medium.
- While waves can move over long distances, the movement of the medium of transmission - the material - is limited.
- So, the oscillating material does not move far from its initial equilibrium position.
- Mechanical waves transport energy which propagates in the same direction as the wave.
- Mechanical waves can be produced only in media which possess elasticity and inertia.

Types of mechanical waves –

- During transmission of a mechanical wave through a medium, the medium particles start to vibrate. On the basis of the direction of particle vibration, mechanical waves are of two types –

(i) Transverse waves

(ii) Longitudinal waves

(i) Transverse Waves :

- A transverse wave is a moving wave that consists of oscillations occurring perpendicular (right angled) to the direction of energy transfer (or the propagation of the wave).

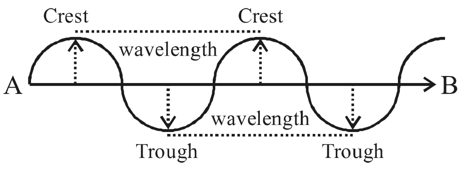
Examples –

a. Vibration in stretched rope : When one end of the rope is bound with hook and another free end is vibrating upward and downward, the produced vibration in rope particle is perpendicular to the wave direction.

b. Waves produced on the water surface.

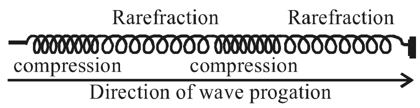
- Transverse waves commonly occur in elastic solids.

- Transverse waves do not originate in gases.
- It originates only on the surface layer of liquids.
- Electromagnetic waves such as light are also transverse waves.
- In transverse wave, maximum displacement in upward side from the equilibrium state is termed as crest while maximum displacement in the downward side is termed as trough. The crest is the top of the wave and trough is the bottom.
- The distance between one wave crest to next wave crest or one trough to next trough is known as wavelength.
- Wavelength is represented by Greek word 'λ'.



(ii) Longitudinal Waves :

- Longitudinal waves are waves in which the displacement of the medium is in the same direction as, or the opposite direction to, the direction of propagation of the wave.
- In longitudinal waves, the oscillations occur in the direction of the wave.
- After stretching if spring is left, longitudinal waves originate in it.
- The places where the circles of spring are very near called compression and the places where circles of spring are far away are called rarefaction.
- The distance between two consecutive compressions or rarefactions is called wavelength of longitudinal waves.



- Longitudinal waves are originated in all mediums i.e. solid, liquid & gas.
- A sound wave (in air and in any fluid medium) is the standard example of a longitudinal wave.
- Along with propagating transverse waves on the surface of the liquid, longitudinal waves can be propagated inside the liquid.
- Longitudinal waves are always mechanical waves.

Time Period :

- In case of a wave propagating in a medium, time taken by particle to complete one vibration is known as the time period of the wave. It is denoted by 'T'.
- Increase in frequency of waves, results into a decrease of time period.

Frequency :

- During propagation of the wave in the medium, frequency is the number of occurrences of repeating event per unit of time by particle.
- It is represented by 'n'.

Relation between Frequency, Speed & Wavelength–

If a vibrating particle is with
Time period – T
Frequency – n, and
Wavelength – λ

then,

Speed of wave (v) = nλ

or speed = frequency × wavelength

since Frequency = $\frac{1}{\text{time period}}$

\ $v = \frac{\lambda}{T}$

2. Electromagnetic Waves :

- Contrast to mechanical waves, some waves need no medium for propagation.
- These waves are called electromagnetic waves.
- Examples of electromagnetic waves are X-rays, light, radio waves etc.
- The speed of all electromagnetic waves are the same and all travel equal speed to the speed of light in a vacuum.
- The electromagnetic spectrum covers electromagnetic waves with frequencies ranging from below one hertz to above 10^{25} hertz, corresponding to wavelengths from thousands of kilometres (10^8m) to a fraction of the size of an atomic nucleus (10^{-14} m).
- Electromagnetic waves with the shortest range of wavelength have higher energy while with longest wavelength range have less energy.

Chart of Electromagnetic Waves

Sl. No.	Name of Wave	Discoverer	Wavelength range	Applications
1.	Gamma Rays	Henry Becquerel Paul Villard	10^{-14} to 10^{-10} m	It has maximum penetrating power. Its application is in nuclear reaction and artificial radioactivity. Gamma rays can kill living cells, they are used to kill cancerous cells. This technique is called Radiotherapy.
2.	X-Rays	Wilhelm Rontgen	10^{-10} to 3×10^{-8} m	Its application is in the field of medical and in industries.
3.	Ultraviolet Rays	Johann Ritter	10^{-8} to 4×10^{-7} m	Hospitals use UV lamps to sterilise surgical equipment and the air in operating theatres. Food & drug companies also use UV lamps to sterilize their products. Suitable doses of ultraviolet rays cause the body to produce Vitamin D.
4.	Visible Radiation (light)	–	4×10^{-7} m to 7.8×10^{-7} m	It is the visible spectrum that is visible to the human eyes and is responsible for the sense of sight.
5.	Infrared Rays	William Herschel	7.8×10^{-7} m to 10^{-3} m	These waves are responsible for heating of any object. These are used in night vision cameras and in TV remote control.
6.	Shortwave Radio or Hertizan waves (Microwaves)	Heinrich Hertz	10^{-3} to 1 m	These are used for the transmission of radio & television signals. The microwaves used in RADAR and microwave oven also use Radio waves. Many celestial objects, such as pulsars emit radio waves.
7.	Longwave Radio	Marconi	1 to 10^{4} m	Applied in the transmission of radio & televisions Programmes.

Question Bank

1. **Which of the following is a mechanical wave?**
   (a) Radio-waves (b) X-rays
   (c) Light waves (d) Sound waves

Uttarakhand P.C.S. (Pre) 2016

Ans. (d)

Sound waves are characterized by the motion of particle in the medium and are longitudinal (in air and in any fluid medium) mechanical waves while Radio-waves, X-rays and light waves are electromagnetic waves.

2. **An example of longitudinal wave is :**
   (a) Radio wave
   (b) Sound wave
   (c) X-ray
   (d) Gamma ray
   (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (b)

See the explanation of above question.

3. **Sound wave in air is –**
   (a) transverse (b) longitudinal
   (c) electromagnetic (d) polarized
   (e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

Sound waves in air (and in any fluid medium) are longitudinal waves because particles of the medium through which sound is transported, vibrate parallel to the direction that the sound wave moves.

4. **What do we call the distance between two consecutive compressions of a sound wave?**
   (a) Wave number (b) Frequency
   (c) Wavelength (d) Amplitude

Jharkhand P.C.S. (Pre) 2023

Ans. (c)

The distance between two consecutive compressions or rarefactions is the wavelength of the sound wave which is a longitudinal wave in air or in any fluid medium. The speed of sound is the product of the wavelength and frequency of the wave. Wavelength in a transverse wave is the distance between two consecutive crests or troughs.

5. **In a Sitar, which type of sound vibrations are produced?**
(a) Progressive and Longitudinal
(b) Progressive and Transverse
(c) Stationary and Longitudinal
(d) Stationary and Transverse
U.P. R.O./A.R.O. (Pre) 2021

Ans. (d)

When we pluck the string of an instrument, like the Sitar, the sound that we hear is not only that of the string. The whole instrument is forced to vibrate, and it is the sound of the vibration of the instrument that we hear. When a string under tension is set into vibration, a transverse wave travels along the wire and is reflected at the fixed end. A transverse stationary wave is thus formed.

6. **Long radio waves are reflected by which of the following layer of Earth's surface –**
(a) Troposphere (b) Ionosphere
(c) Tropopause (d) Stratosphere
U.P. P.C.S. (Pre) 1991

Ans. (b)

The ionosphere is a region of Earth's upper atmosphere from about 50 to 400 miles altitude. It is ionized by solar radiation. It has practical importance because among other functions, it influences radio propagation to distant places on Earth.

7. **Wireless communication is reflected back to the earth's surface by the –**
(a) Troposphere (b) Stratosphere
(c) Ionosphere (d) Exosphere
U.P.P.C.S. (Pre) 1998

Ans. (c)

See the explanation of above question.

8. **Which of the following atmospheric layers is responsible for the deflection of radiowaves ?**
(a) Troposphere (b) Stratosphere
(c) Mesosphere (d) Ionosphere
U.P. Lower Sub. (Pre) 1998

Ans. (d)

See the explanation of above question.

9. **Waves of the Ultra High Frequency (UHF) range normally propagate by means of**
(a) Ground waves (b) Sky waves
(c) Space waves (d) Surface waves
R.A.S. / R.T.S. (Pre) 2018

Ans. (c)

Ultra high frequency (UHF) is the ITU designation for radio frequencies in the range between 300 MHz and 3 GHz. Owing to its high frequency, an ultra-high frequency (UHF) wave can neither travel along the trajectory of the ground nor get reflected by the ionosphere. The signals having UHF are propagated normally through the line of sight communication which is actually space wave propagation. The radio waves having high frequencies are basically called as space waves.

10. **Which of the following cannot travel in vacuum?**
(a) Light (b) Heat
(c) Sound (d) Electromagnetic waves
M.P.P.C.S. (Pre) 2000
Jharkhand P.C.S. (Pre) 2023

Ans. (c)

Sound waves cannot travel in vacuum. It is transmitted by the movement of particles along with the direction of the motion of the sound wave. More generally, sound is a mechanical disturbance which is dependent upon a medium to travel. It can be transmitted through solid, liquid, and gas.

11. **Assertion (A): Radio waves bend in a magnetic field.**
Reason (R) : Radio waves are electromagnetic in nature.
Code :
(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.
I.A.S. (Pre) 2008

Ans. (d)

Assertion (A) is false but reason (R) is correct because radio waves are electromagnetic in nature. So they are generally unaffected by magnetic and electric field.

12. **Cosmic rays –**
(a) Are charged particles
(b) Are uncharged particles
(c) Can be charged as well as uncharged
(d) None of the above
56th to 59th B.P.S.C. (Pre) 2015

Ans. (a)

Cosmic rays are not the part of electromagnetic spectrum. They are immensely high-energy charged particles, travelling through space at a speed approaching that of light. They originated either from the sun or outside of our solar system.

13. **Which one of the following statements is not true about cosmic rays?**
(a) They are electromagnetic waves.
(b) They have very short wavelength.
(c) They are made of highly energetic charged particles.
(d) They originate from the Sun

U.P.P.C.S. (Pre) 2005

Ans. (a)

See the explanation of above question.

14. **Which of the following does not require a medium?**
(a) Radiation (b) Convection
(c) Conduction (d) None of these

U.P. R.O./A.R.O. (Mains) 2017

Ans. (a)

In physics, radiation is the emission or transmission of energy in the form of waves or particles through space or through a medium. Electromagnetic radiations such as radio waves, microwaves, infrared, visible light, X-rays and gamma rays do not require a medium. On the other hand heat transmission by convection and conduction requires a medium.

15. **Which one of the following is associated with 'Albedo'?**
(a) Transmitting power (b) Absorbing power
(c) Emissive power (d) Reflecting power

U.P.P.C.S. (Pre) 2019

Ans. (d)

Albedo is the fraction of light or radiation that is reflected by a body or surface. It is commonly used in astronomy to describe the reflective properties of planets, satellites, and asteroids. The range of albedo on the Earth's surface can be as high as 95% (0.95) for fresh snow cover and as little as 3% (0.03) for water.

16. **Consider the following statements:**
Statement-I : The atmosphere is heated more by incoming solar radiation than by terrestrial radiation.
Statement-II : Carbon dioxide and other greenhouse gases in the atmosphere are good absorbers of long wave radiation.
Which one of the following is correct in respect of the above statements?
(a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
(b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
(c) Statement-I is correct, but Statement-II is incorrect
(d) Statement-I is incorrect, but Statement-II is correct

I.A.S. (Pre) 2024

Ans. (d)

The Earth's surface receives most of its energy from the Sun in short wavelengths. The energy received by the Earth is known as incoming solar radiation which in short is termed as insolation. The atmosphere is largely transparent to short wave solar radiation. The incoming solar radiation passes through the atmosphere before striking the Earth's surface. Within the troposphere water vapour, ozone and other gases absorb much of the near infrared radiation.
The insolation received by the Earth is in short waves forms and heats up its surface. The Earth after being heated itself becomes a radiating body and it radiates energy to the atmosphere in long wave form. This energy heats up the atmosphere from below. This process is known as terrestrial radiation. The long wave radiation is absorbed by the atmospheric gases particularly by carbon dioxide and the other greenhouse gases. As a whole (as per the NCERT), about 48 percent of incoming Sun's energy is absorbed by the atmosphere, in which about 14 units absorbed directly from insolation while about 34 units indirectly from terrestrial radiation. Thus, the atmosphere is heated more indirectly by the terrestrial radiation, and not directly by the insolation. Hence, Statement-I is incorrect, but Statement-II is correct.

17. **The Earth's atmosphere is mainly heated by which one of the following?**
(a) Scattered solar radiation
(b) Shortwave solar radiation
(c) Longwave terrestrial radiation
(d) Reflected solar radiation

U.P. P.C.S. (Pre) 2022

Ans. (c)

The energy released from the Sun is emitted as shortwave light and ultraviolet energy. When it reaches the Earth, some is reflected back to space (about 35%), some is absorbed by the atmosphere (about 14%), and some is absorbed in the Earth's surface (about 51%). The insolation received by the Earth is in shortwave form and heats up its surface. The Earth after being heated itself becomes a radiating body and it radiates energy to the atmosphere in longwave form. This energy heats up the atmosphere from below. This process is known as terrestrial radiation. The longwave terrestrial radiation is absorbed by the atmospheric gases particularly by carbon dioxide and the other greenhouse gases. Hence, the Earth's atmosphere is mainly heated by the longwave terrestrial radiation.

18. **The atmosphere is mainly heated by the :**
(a) long-wave terrestrial radiation
(b) short-wave solar radiation

(c) reflected solar radiation
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

See the explanation of above question.

19. Consider the following statements and choose the right options given below.
(1) Earth's surface re-emits heat in the form of Ultra-violet radiation.
(2) Clouds and gases reflect about one fourth of the incoming solar radiation.
(a) Only the statement (1) is correct
(b) Only the statement (2) is correct
(c) Both the statements (1) and (2) are incorrect
(d) Both the statements (1) and (2) are correct

Chhattisgarh P.C.S. (Pre) 2023

Ans. (b)

As per the NCERT, clouds and gases reflect about one-fourth of the incoming solar radiation, and absorb some of it but almost half of incoming solar radiation falls on Earth's surface heating it, while a small proportion is reflected back. Earth's surface re-emits heat in the form of infrared radiation but part of this does not escape into space as atmospheric gases (e.g., carbon dioxide, methane, etc.) absorb a major fraction of it. Hence, only statement (2) is correct. However, as per some other sources clouds and gases reflect about 17 – 20 percent of the incoming solar radiation. This may be the reason that CGPSC has given option (c) as the answer in its final answer key.

20. What is the distance between two successive crests or successive troughs called?
(a) Amplitude (b) Wavelength
(c) Frequency (d) None of these

45th B.P.S.C. (Pre) 2002

Ans. (b)

Wavelength means the distance measured in the direction of a wave from any given point to the next point in the same phase, as from crest to crest. The distance between two successive crests or two successive troughs is the wavelength of a transverse wave.

21. The velocity of electromagnetic waves is :
(a) 3×10^8 ms^{-1} (b) 3×10^7 ms^{-1}
(c) 3×10^6 ms^{-1} (d) 3×10^5 ms^{-1}
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (a)

The velocity of all electromagnetic waves is the same and all travel at equal speed to the speed of light in a vacuum. Hence, the velocity of electromagnetic waves is about 3×10^8 ms^{-1}.

22. One of the properties associated with X-rays is that they can be deflected by :
(a) Both electric and magnetic fields together
(b) Electric fields only
(c) Magnetic fields only
(d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (d)

X-rays are electromagnetic waves or radiation and do not carry an electric charge, therefore they are not deflected by electric or magnetic fields. They can be diffracted or scattered by matter, particularly by electrons within an atom. This interaction is the basis of X-ray diffraction and imaging techniques.

23. Which one is not an example of electromagnetic wave?
(a) g-rays (b) X-rays
(c) Ultraviolet rays (d) Supersonic waves

Jharkhand P.C.S. (Pre) 2021

Ans. (d)

Among the given options, supersonic waves are not an example of electromagnetic waves, while g (Gamma)-rays, X-rays and Ultraviolet rays are examples of supersonic waves. Supersonic is used for objects which travel at a speed greater than the speed of sound.

24. Which of the following has the longest wavelength ?
(a) Infrared (b) X-rays
(c) Visible light (d) Radio waves

R.A.S./R.T.S. (Pre) 1996

Ans. (d)

The electromagnetic spectrum consists of all the different wavelength of electromagnetic radiations such as: Radio waves > Microwave > Infrared > Visible > Ultraviolet > X-rays > Gamma rays. Thus it is clear that the radio waves are having the maximum wavelength while the Gamma rays are having minimum wavelength.

25. Which of the following electromagnetic radiations has the maximum energy?
(a) Visible light (b) Infrared rays
(c) Ultraviolet rays (d) X-rays

U.P. P.C.S. (Pre) 2018
U.P. P.C.S. (Mains) 2005

Ans. (d)

Shorter the wavelength, greater is the frequency and energy. In reference to energy, the sequence of the energy of electromagnetic rays is as follows :
Radio waves < Microwaves < Infrared < Visible light < Ultraviolet rays < X-rays < Gamma rays.
It is clear that among the given options, the X-rays have the maximum amount of energy than others.

26. Consider the following and arrange these in increasing order of frequency :

I. X-rays **II. Visible rays**

III. Infrared rays **IV. Radio waves**

Select the correct answer using the code given below :

Code :

(a) III, II, I, IV (b) I, II, III, IV

(c) II, III, IV, I (d) IV, III, II, I

U.P. R.O./A.R.O. (Pre) 2023

Ans. (d)

The correct increasing order of the given electromagnetic waves in terms of frequency is as follows :
Radio waves < Infrared rays < Visible rays < X-rays.

27. Microwaves are electromagnetic waves having frequencies in range of :

(a) 300 KHz – 3 MHz (b) 3 MHz – 300 MHz

(c) 1 GHz – 300 GHz (d) 300 GHz – 400 THz

M.P. P.C.S. (Pre) 2020

Ans. (c)

Microwave is a form of electromagnetic radiation with wavelengths ranging from about one meter to one millimeter corresponding to frequencies between 300 MHz and 300 GHz respectively. Different sources define different frequency ranges as microwave; the above broad definition includes both UHF and EHF (millimeter wave) bands. Hence, option (c) is the most appropriate answer.

28. Which one of the following is used for determining the structure of crystal :

(a) Gamma rays (b) X-rays

(c) UV rays (d) visible lights

R.A.S./R.T.S. (Pre) 1997-98

Ans. (b)

Crystallography is the science that examines crystals which can be found everywhere in nature, from salt to snowflakes to gemstones. Crystallographers use the properties of the inner structure of crystals to determine the arrangement of atoms and generate knowledge which is used by chemist, physicists and other. Crystallographers use X-ray, neutron, and electron diffraction techniques to identify the characteristics of solid materials.

29. A radar which detects the presence of an enemy aircraft uses :

(a) Light waves (b) Radio waves

(c) Sound waves (d) Ultrasound waves

Uttarakhand P.C.S. (Mains) 2002

Ans. (b)

Radar is an object- detection system which uses radio waves to determine the range, angle or velocity of objects. It is used to detect the location of aircraft, ships, spacecraft, motor vehicle etc.

30. Remote Sensing uses which of the following in its applications?

(a) Electric waves (b) Sonar waves

(c) Gamma rays (d) Electromagnetic waves

Uttarakhand P.C.S. (Pre) 2021

Ans. (d)

Remote sensing is the process of detecting and monitoring the physical characteristics of an area by measuring its reflected and emitted radiation at a distance (typically from a satellite or aircraft). Remote sensing uses a part or several parts of the electromagnetic spectrum. It records the electromagnetic energy reflected or emitted by the Earth's surface. When electromagnetic radiation hits the surface of an object, different wavelengths are either reflected or absorbed depending on the physical and chemical properties of the object. Special cameras collect remotely sensed images, which help researchers 'sense' things about the Earth.
It is to be noted that gamma rays are part of this electromagnetic radiation and are also used in remote sensing (in spacecrafts exploring the Moon or other planets but not in typical remote sensing of Earth observation satellites as this part of radiation is absorbed by the Earth's atmosphere). Remote sensing observation using gamma ray spectroscopy is an effective technique to determine the elemental composition of planetary surface. Gamma ray remote sensing observations find important applications in the study of the development of the planets.

31. When there is depletion of ozone in the stratosphere, the wavelength of radiation striking the Earth's surface will be –

(a) 10^{-10} m (b) 10^{-7} m

(c) 10^{-2} m (d) 100 m

I.A.S. (Pre) 1993

Ans. (b)

Ultraviolet radiations are mainly divided into three groups:-

UV-A radiations : The long wave UV-A radiations having the wavelength of 320-400 nm. They strike the surface of the Earth as the part of the rays of the sun.

UV-B radiations : The medium wave UV-B radiation has the wavelength of 280-320 nm. It is mostly absorbed by the ozone layer, but some do reach the Earth's surface.

UV-C radiations : It has a wavelength of 100-280 nm. It is completely absorbed by ozone layer and atmosphere.

Therefore, on depletion of ozone in the stratosphere, the minimum wavelength of radiation striking the surface of the Earth will be of 100 nm.

$$1 \text{ nm} = 1.0 \times 10^{-9} \text{ m}$$
$$100 \text{ nm} = 100 \times 10^{-9}$$
$$= 10^2 \times 10^{-9} = 10^{-7} \text{ m}$$

32. Which one of the following types of waves are used in a Night Vision apparatus?

(a) Radio waves (b) Microwaves
(c) Infrared waves (d) None of these

I.A.S. (Pre) 2009

Ans. (c)

Infrared waves are the type of electromagnetic radiations with longer wavelengths compared to those of visible light. It is used in night vision equipment when there is insufficient visible light to see. It is used by the soldiers to find the target, intruders and hidden bombs in night, thus making the application of force more discriminating.

33. The waves used in common TV remote control are

(a) X-Rays (b) Ultraviolet Rays
(c) Infrared Rays (d) Gamma Rays

R.A.S./R.T.S. (Pre) 2018

Ans. (c)

Infrared rays are commonly used in TV remote control. Remote control can be used to operate devices such as television set, DVD player or other home appliances.

34. Which type of electromagnetic radiation is used in the remote control of a television?

(a) Infrared (b) Ultraviolet
(c) Visible (d) None of the above

U.P.P.C.S.(Pre) 2013
U.P. U.D.A./L.D.A. (Pre) 2010
U.P.P.C.S (Pre) 2010
U.P.P.C.S. (Pre) 2002

Ans. (a)

See the explanation of above question.

35. Which electromagnetic radiation is used in remote control of a television?

(a) Infrared (b) Ultraviolet
(c) Microwave (d) None of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (a)

See the explanation of above question.

36. Which one of the following does a TV remote control unit use to operate a TV set ?

(a) Light waves (b) Sound saves
(c) Microwaves (d) Radio waves

U.P.P.C.S. (Mains) 2013
I.A.S. (Pre) 2000

Ans. (d)

Infrared waves are generally used in a TV remote control unit to operate a TV set, but some of them use radio waves. Infrared radiation and radio waves both are electromagnetic radiation.

37. Waves used for telecommunication are –

(a) Visible light (b) Infrared
(c) Ultraviolet (d) Microwave

U.P.P.C.S. (Mains) 2013

Ans. (d)

The microwaves are high-frequency signals in the 300 MHz to 300 GHz range. The signals can carry thousands of channels at the same time, making it a very versatile communication system. Microwave is often used for point-to-point telecommunications. Today microwave is employed by telecommunication industry in the form of both terrestrial relays and satellite communication.

38. The process of superposition of any information on radiowaves is named as :

(a) transmission (b) modulation
(c) demodulation (d) reception

Jharkhand P.C.S. (Pre) 2023

Ans. (b)

The process of superimposing any information (audio or video messages) on carrier radiowaves is called modulation.

39. What is not true about FM (Frequency Modulation) radio waves?

(a) Frequency range of Frequency Modulation is lower than that of Amplitude Modulation
(b) Amplitude and phase do not change in it
(c) It was invented by Edwin Armstrong
(d) Its bandwidth is higher than that of Amplitude Modulation

Chhattisgarh P.C.S. (Pre) 2021

Ans. (a)

Edwin Howard Armstrong was an American electrical engineer who invented wideband frequency modulation (FM) radio. Armstrong presented his paper, "A Method of Reducing Disturbances in Radio Signaling by a System of Frequency Modulation", (which first described FM radio) before the New York section of the Institute of Radio Engineers on November 6, 1935. The paper was published in 1936. FM requires a wider signal bandwidth (and higher frequency range) than amplitude modulation (AM) by an equivalent modulating signal; this also makes the signal more robust against noise and interference. In frequency modulation, the carrier amplitude and phase remains constant, but its frequency changes in accordance with the modulating signal.

40. Which modulation technique is preferred for medium speed communication range upto 1200 to 2400 bits per second?

(a) Amplitude modulation (b) Frequency modulation
(c) Phase modulation (d) None of these

Uttarakhand P.C.S. (Pre) 2024

Ans. (b)

Frequency modulation (FM) technique is preferred for medium speed communication range upto 1200 to 2400 bits per second. Amplitude modulation (AM) radio frequency ranges from 530 to 1710 KHz or at max 1200 bits per second while FM radio ranges in a higher spectrum from 88 to 108 MHz or 1200 to 2400 bits per second.

41. FM broadcasting service uses the range of frequency bands between –

(a) 109-139 MHz (b) 54-75 MHz
(c) 76-87 MHz (d) 88-108 MHz

R.A.S/R.T.S. (Pre) 2013

Ans. (d)

See the explanation of above question.

42. A radio station broadcast at 30 metre band. The frequency of the carrier wave transmitted by this station is:-

(a) 10 KHz (b) 100 KHz
(c) 10 MHz (d) 100 MHz

R.A.S./R.T.S. (Pre) 1993

Ans. (c)

According to formula $n = \frac{c}{I}$

[where n = frequency, c = speed of light; I = wavelength]

$$n = \frac{3 \times 10^8}{30} = 10 \times 10^6 = 10\text{MHz}$$

43. A layer in the Earth's atmosphere called Ionosphere facilitates radio communication. Why?

1. The presence of ozone causes the reflection of radio waves to Earth.

2. Radio waves have a very long wavelength.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2011

Ans. (d)

Both the statements are incorrect in the context of the question. Radio waves have a very long wavelength but the presence of electrically charged ions in ionosphere facilitate radio communication.

44. Television viewers using dish antenna to receive satellite signals do not receive signals during rain because :

1. of small size of antenna.

2. rain droplets absorb the energy of radio waves.

3. rain droplets disperse the energy of radio waves from their original direction.

Which of the above statements are correct?

(a) only 1 (b) only 1 and 2
(c) only 2 and 3 (d) 1, 2, and 3

U.P.P.C.S. (Pre) 2017

Ans. (d)

During rain, radio waves collide with raindrops partly or completely and converted into thermal energy absorbing the energy of radio waves. Rain drops are also capable of disturbing the basic direction of energy of the radio waves. Due to this reason, there is difficulty in receiving satellite signals during rain. Radio signals become weak during rain which can not be received by small antenna.

45. Following rays are used in the diagnosis of intestinal diseases :

(a) X-rays (b) α-rays
(c) β-rays (d) γ-rays

43rd B.P.S.C. (Pre) 1999

Ans. (a)

X-rays are a form of electromagnetic radiations, which are used in the diagnosis of intestinal diseases.

46. CT Scan is done by using –

(a) Infrared Rays (b) Ultrasonic Waves
(c) Visible Light (d) X-Rays

U.P.P.C.S. (Mains) 2011

Ans. (d)